Assignment 1

1) If $\bar{D} = (2y^2+z)\,\bar{a}_x + 4xy\,\bar{a}_y + x\,\bar{a}_z$ C/m^2. Find
   a) volume charge density at $(-1, 0, 3)$
   b) The flux through the cube defined by
   $0 \le x \le 1$, $0 \le y \le 1$, $0 \le z \le 1$.
   c) The total charge enclosed by the cube.

→

Given $\bar{D} = (2y^2+z)\,\bar{a}_x + 4xy\,\bar{a}_y + x\,\bar{a}_z$

i) $\nabla \cdot \bar{D} = \rho_v$

$$\frac{\partial}{\partial x}(2y^2+z) + \frac{\partial}{\partial y}(4xy) + \frac{\partial}{\partial z}\, x$$

$$\rho_v = 4x$$

$$\rho_v(-1, 0, 3) = -4 \quad C/m^3$$

ii) $$\phi = \iiint_v \nabla \cdot \bar{D}\, dv = \iiint_v 4x\, dv$$

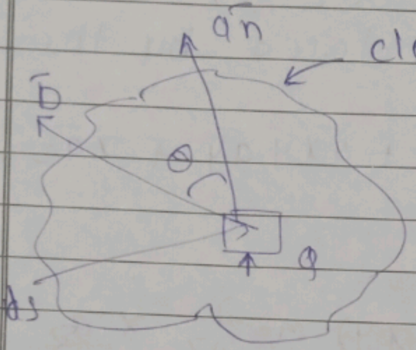
$$= \int_{x=0}^{1} \int_{y=0}^{1} \int_{z=0}^{1} 4x \, dz\, dy\, dx$$

$$= 4\left[\frac{x^2}{2}\right]_0^1 = 4 \cdot \frac{1}{2} = \underline{\underline{2}} \quad C$$

iii) $$Q = \phi = \iiint_v \rho_v \, dv = \underline{\underline{2}} \quad C$$

Q.2) a) State and explain Gauss law for electrostatics?

- The electric flux passing through any closed surface is equal to the charge enclosed by that surface.



closed irregular surface.

- consider integral closed surface in which total charge enclosed is Q C;

- Therefore, according to faraday's experiment total flux that has to be passed through the closed surface is Q.
- consider a diff. surface area ds, then we can write

$$d\bar{S} = ds\,\bar{a}_n$$

where $\bar{a}_n$ is equal to normal to the surface ds at pt. given.

$$d\psi = D_n ds \quad \text{—— (1)}$$

$$D_n = |\bar{D}| \cos\theta$$

eq^n (1) becomes.

$$d\psi = |\bar{D}| \cos\theta \; ds \quad \text{— (2)}$$

By using the def^n of dot product the

above eqn can be written as.

$$|\bar{D}| \cos\theta \, ds = \bar{D} \cdot ds$$

$$d\psi = \bar{D} \cdot d\bar{s} \quad \text{—— (3)}$$

- This is the flux passing through the diff. area ds. Hence the total flux passing through the entire close surface is obtained by

$$\psi = \oint d\psi = \oint_s \bar{D} \cdot d\bar{s}$$

$$Q = \psi = \oint_s \bar{D} \cdot d\bar{s}$$

b) Explain what is potential gradient. State and prove divergence theorem.

- Potential gradient is a vector quantity that shows how scalar potential field changes in space.

$$\nabla . V = \frac{\partial v}{\partial x} \bar{a_x} + \frac{\partial v}{\partial y} \bar{a_y} + \frac{\partial v}{\partial z} \bar{a_z}$$

- Divergence thm
If any close surface has charge q and electric flux density $\bar{D}$ then according to gauss law

$$q = \oint_s \bar{D} \cdot d\bar{s} \quad \text{—— (1)}$$

- If any volume charge distribution has volume charge ρ_v, then charge enclosed by same volume charge distribution is given by

$$Q = \int_v \rho_v \, dv \quad \text{--- (2)}$$

from (1) & (2)

$$\oint_S \bar{D} \cdot d\bar{S} = \int_v \rho_v \, dv \quad \text{--- (3)}$$

But $\rho_v = \nabla \cdot \bar{D}$ eqn (3) $\Rightarrow$

$$\oint_S \bar{D} \cdot d\bar{S} = \int_v (\nabla \cdot \bar{D}) \, dv \quad \text{--- (4)}$$

Thus divergence thm may be defined as surface integration of any flux density equal to volume integration of diversions of same electric field.

Timely submission	understan-ding	punctuality	total

Assignment 2

1) a) State and explain Gauss's law for magnetostatics ?

→ - The total magnetic flux through any closed surface is always zero.

$$\oint_S \bar{B} \cdot d\bar{s} = 0$$

$\bar{B}$ = magnetic flux density

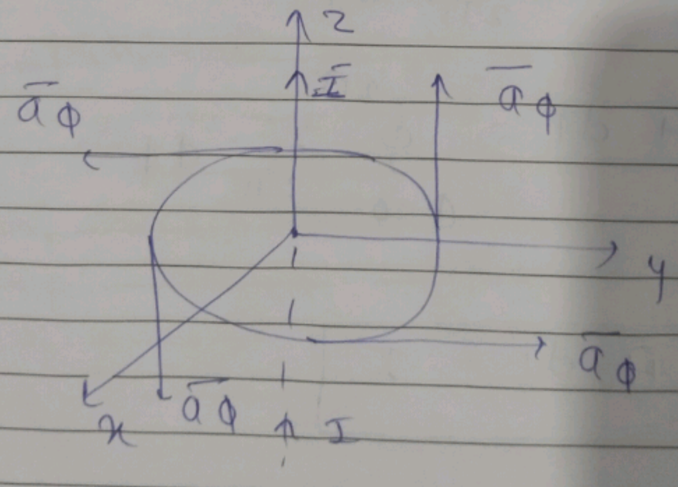
$d\bar{s}$ = small area vector normal to the surface s.

$\oint_S$ = surface integral over a closed surface.

b) State and prove Ampere circuital law.

→ - ACL states that the line integral of $\bar{H}$ around closed path is the same as the net current enclosed by the path.

$$I = \oint \bar{H} \cdot dt \quad - (1)$$



- consider a long straight conductor carrying current I placed along z-axis.
- consider a close circular path of radius ρ which enclosed the straight conductor.
- The pt. P is a $\perp$er distance ρ from the conductor.

$$d\bar{l} = \rho \, d\phi \, \bar{a}_\phi \quad \text{— (2)}$$

we know that,

$$\bar{H} = \frac{I}{2\pi\rho} \bar{a}_\phi$$

$$\bar{H} d\bar{l} = \frac{I}{2\pi\rho} \bar{a}_\phi \cdot \rho \, d\phi \, \bar{a}_\phi$$

$$\bar{H} d\bar{l} = \frac{I}{2\pi} d\phi \qquad (\bar{a}_\phi \cdot \bar{a}_\phi = 1)$$

$$\oint \bar{H} \cdot d\bar{l} = \oint_{\phi=0}^{2\pi} \frac{I}{2\pi} d\phi$$

$$\boxed{\oint \bar{H} \cdot dl = I}$$

2) a) State maxwell's equation for static field in integral and point form.

It is given by

$$\nabla \times \bar{H} = \bar{J}$$

It is also known as point form of ACL

$$\nabla \times \bar{H} = \begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix}$$

In cylindrical

$$\nabla \times \bar{H} = \frac{1}{\rho} \begin{vmatrix} \bar{a}_\rho & \rho \bar{a}_\phi & \bar{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ H_\rho & \rho H_\phi & H_z \end{vmatrix}$$

In spherical

$$\nabla \times \bar{H} = \frac{1}{r^2 \sin\theta} \begin{vmatrix} \bar{a}_r & \bar{r} a_\theta & r\sin\theta\, \bar{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ H_r & \partial H_\theta & r\sin\theta\, \bar{a}_\phi \end{vmatrix}$$

b) Define conduction current and conduction current density.

- conduction current (I)
The flow of electric charge through a conductor per unit time.
measured in A.

$$I = \int_S J \, ds$$

- conduction current density (J)
A vector Field giving current per unit area (A/m^2)
Direction is local direction of +ve charge flow; mag. is current per unit area through a tiny surface normal to J.

Timely submission	under-standing	Punctuality	total

Assignment 6

1) A) Explain the primary line constants (R, L, G, C) and secondary constants (Z_0, γ, α, β) of transmission line in details also derive relation betn primary constant and secondary constant of transmission line.

- Primary line constants

1) R (Resistance per unit length)
- Represents the ohmic loss due to resistance of conductor.
- unit : Ω/km

2) L (Inductance per unit length)
- Represents magnetic field energ storage
- unit : H/km

3) G (conductance)
- Represents leakage current through dielectric between conductors.
- unit : S/km

4) (Capacitance) C
- Represents electric field energy storage betn conductors.
- unit : F/km

- Secondary line constants.

1) Z_0 (characteristic impedance)
- Ratio of voltage to current of a

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wave travelling along the line.

2) γ (propagation constant)
Describes how signal amplitude and phase change along the line.

3) α (attenuation constant)
Represents signal amplitude decay due to losses. unit : n/km

4) β (phase constant)
Represents phase shift of signal per unit length.
unit rad/km.

• Relation

$$Z_0 = \sqrt{\frac{Z}{Y}} \qquad \frac{\text{Impedance}}{\text{Admittance}}$$

$$Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$$

$$\gamma = \sqrt{ZY} = \sqrt{(R+j\omega L)(G+j\omega C)}$$

B) A transmission line cable has following primary constants $R = 11$ ohm/km, $G = 0.8\ \mu\mho$/km $L = 0.00367$ H/km $C = 8.35$ nF/km at a signal of 1kHz

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calculate.

1) characteristic impedence Z_0

2) α

3) β

4) λ

5) v

→ Given

$R = 11 \ \Omega/km$

$G = 0.8 \times 10^{-6} \ \mho/km$

$L = 0.00367 \ H/km$

$C = 8.35 \times 10^{-9} \ F/km$

$F = 1 \ kHz$

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \qquad \omega = 2\pi f, \quad \omega = 2\pi \times 10^3$$

$$= \sqrt{\frac{11 + j\,2\pi \times 10^3 \times 0.00367}{0.8 \times 10^{-6} + j\,2\pi \times 10^3 \times 8.35 \times 10^{-9}}}$$

$$= \sqrt{\frac{25.548 \angle 64.49}{5.247 \times 10^{-5} \angle 89.12}}$$

$$= \sqrt{486906.80 \angle -24.63}$$

$$Z_0 = 697.78 \angle -12.315^\circ$$

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$\gamma = \sqrt{(25.548 \angle 64.49)(5.24 \times 10^{-5} \angle 89.12)}$

$\gamma = \sqrt{1.338 \times 10^{-3} \angle 153.61}$

$\gamma = 0.03657 \angle 76.805$

~~$\alpha = 0.03657$~~
~~$\beta =$~~

$\gamma = 8.3476 \times 10^{-3} + j\,0.0356$

$\alpha = 8.3476 \times 10^{-3}$
$\beta = 0.0356$

$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{0.0356} = 176.25$ km

$v = \frac{\omega}{\beta} = 176493.98$ km/s

2) A) A 50 Ω transmission line is terminated in a load $Z = 25 + j50\ \Omega$. the length of transmission line is 3.3λ. find the following using smith chart.
i) VSWR
ii) Reflection coefficient
iii) Input impedence

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iv) Input admittance.

→ A) Using Smith chart

$Z_L = 25 + j50\ \Omega$

$Z_0 = 50\ \Omega$

Z_L = normalizing = ~~60+~~ $\frac{25+j50}{50}$

$Z_L = 0.5 + 1j$

$R = 0.5$

$X = 1$

ii) $|K| = \frac{OP}{OQ} = \frac{4.7}{7.6} = 0.618$

$\angle K = 82^\circ$

$S = 4.3$

$l = 3.3\lambda$

$l = \frac{30 \times \cancel{3.3} \times 360 \times 2}{3.3} =$

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theoritical

$$K = \frac{Z_R - Z_0}{Z_R + Z_0} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$= \frac{25 + j50 - 50}{25 + j50 + 50} = \frac{-25 + j50}{75 + j50}$$

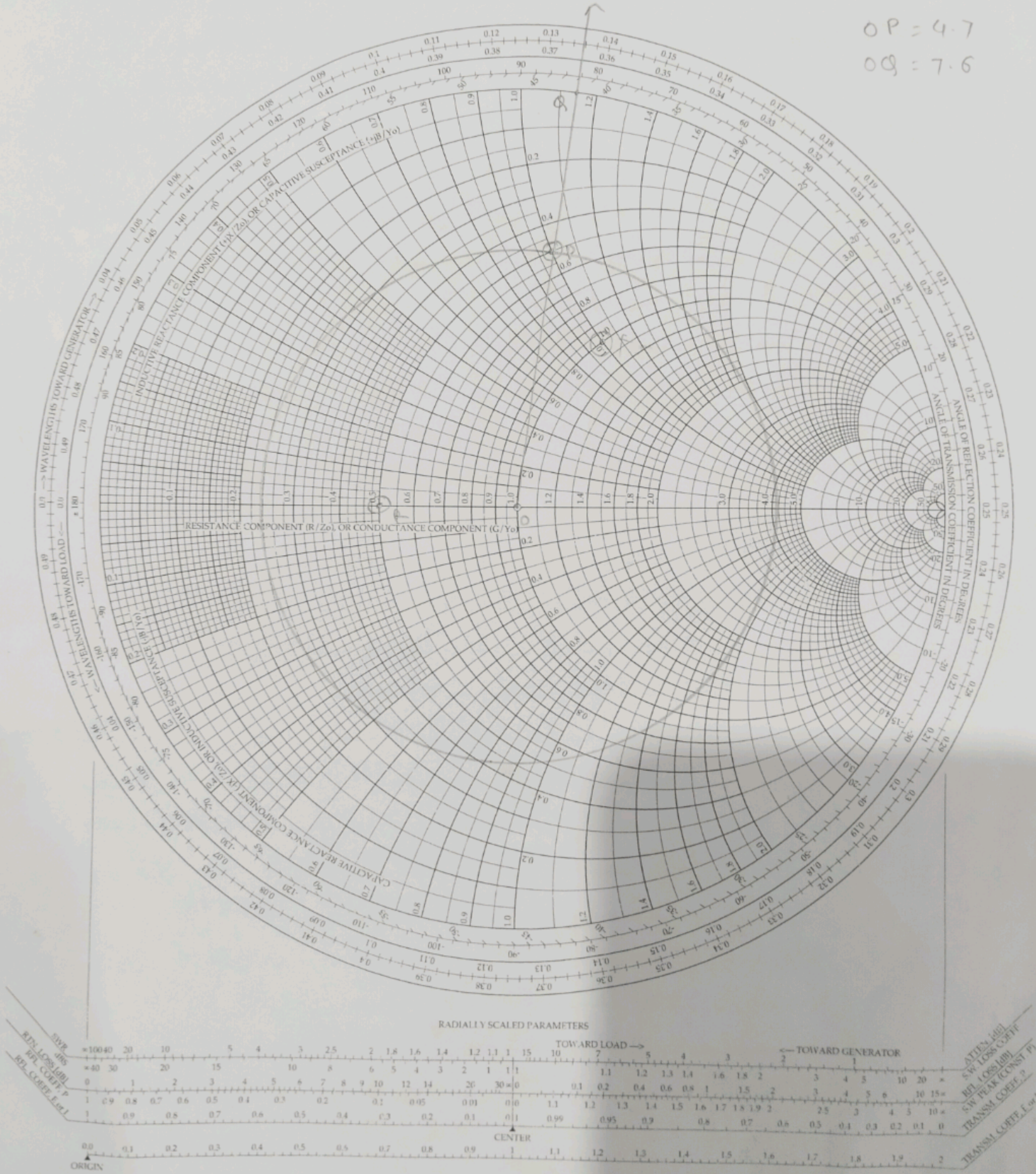
$$K = \frac{55.90 \angle 116.56}{90.13 \angle 33.69} = 0.6202 \angle 82.87$$

$$S = \frac{1 + |K|}{1 - |K|} = 4.26$$

The Complete Smith Chart

Black Magic Design

OP = 4.7
OQ = 7.6



B) Explain different distortions of transmission lines? What is mean by distonless line and explain the codition of distortion less line.

→ There are two types of distortions

i) Attenuation distortion

- Occurs when different frequency components of signal are attenuated unequally during transmission.
- In a complex waveforms, there are many harmonics. If high frequencies are attenuated more than low frequencies the signal shap changes.
- causes loss of signal strength and change in waveform shape.
- unequal resistance & conductance with freq. cause varying attenuation.

ii) phase (Delay) Distortion.

- occurs when different freq. components of a s/g experience diff. phase shifts during transmission.
- Each freq. travels at a different velocity, so they reach the receiving end at different times.
- The waveforms shape changes even if amplitudes are constant.
- variation of phase constant (β) or velocity of propagation with freq.

- Distortionless transmission line
- A transmission line is said to be distortionless if the waveform of s/g is not changed during transmission - only its amplitude may decrease uniformly.

condition for distortionless line

$$\frac{R}{L} = \frac{G}{C}$$

$$LG = RC$$